exercise2.py

```
from sklearn.feature_extraction.text import TfidfVectorizer

# Collect user input
documents = []
text = input("Enter document contents or type `exit` to end
while text.strip() != 'exit':
    documents.append(text.strip())
    text = input("Enter document contents or type `exit` to

# WRITE YOUR CODE HERE
vectorizer = TfidfVectorizer(stop_words='english')
X = vectorizer.fit_transform(documents)
print(vectorizer.get_feature_names())
print(X.toarray())
```

100% (14:19)

Terminal

```
codio@horizonliquid-elasticsusan:~/workspace$ python3 exercise2.py
Enter document contents or type `exit` to end input: This is the first
Enter document contents or type `exit` to end input: This document is t
Enter document contents or type `exit` to end input: And this is the th
Enter document contents or type `exit` to end input: Is this the first
Enter document contents or type `exit` to end input: exit
/home/codio/.local/lib/python3.9/site-packages/sklearn/utils/deprecatio
et_feature_names is deprecated; get_feature_names is deprecated in 1.0
 use get_feature_names_out instead.
  warnings.warn(msg, category=FutureWarning)
['document', 'second']
[[1.         0.        ]
 [0.78722298 0.61666846]
 [0.         0.        ]
 [1.         0.        ]]
codio@horizonliquid-elasticsusan:~/workspace$
```

Guide

Collapse

Coding Exercise 2

In the top-left pane, implement a TF-IDF scored bag of words model using `documents` which will capture keyboard input.

When creating your model, make sure to:

- Use `TfidfVectorizer`
- Filter out punctuation
- Filter out stop words
- Ignore case

Output the (1) vocabulary and the (2) vectors.

Test your code using the terminal in the bottom-left:

TRY IT

Example Input/Output:

```
This is the first document.
This document is the second document.
And this is the third one.
Is this the first document?
exit

['document', 'second']
[[1.         0.        ]
 [0.78722298 0.61666846]
 [0.         0.        ]
 [1.         0.        ]]
```

Submit your code for feedback by pressing the button below.

A solution:

```
vectorizer = TfidfVectorizer(stop_words='e
X = vectorizer.fit_transform(documents)

print(vectorizer.get_feature_names())
print(X.toarray())
```

Note that many of the design constraints are handled automatically – but you do need to specify that stop words are filtered out.

Check It!

LAST RUN on 12/27/2023, 4:43:23 AM

Check 1 passed

Check 2 passed

Next